Leonid Vinichenko

Tech Lead, Expert 3D Graphics Software Engineer, Fullstack (TypeScript / C++)

Email: vinneyto@gmail.com
Phone: +37443751861
Yerevan

Telegram: @vinneyto GitHub: vinneyto

Summary: Developer with 10+ years of experience in frontend and 3D graphics (WebGL/WebGPU) in the browser. Currently Tech Lead of a small team, working as a fullstack engineer in TypeScript and C++. Responsible for architectural decisions, early architectural reviews (with a focus on SOLID), feature design and decomposition across frontend, backend, and design, and delivery to production.

Work Experience

Tech Lead / Expert 3D Graphics Software Engineer at Align Technology

February 2024 – Current

Tech Lead of a team of 2 engineers, developer and architect of a 3D system for a web app used by orthodontists. Work as a fullstack engineer across **TypeScript** (frontend) and **C++** (backend). Own features end-to-end: from design and decomposition to delivery in production.

- Lead a team of 2 engineers: planning, code reviews, mentoring, and day-to-day technical guidance
- Design features and drive them to production: align requirements with stakeholders and decompose work across **frontend**, **backend**, and **design**
- Work as a fullstack engineer, writing code in both **TypeScript** (frontend) and **C++** (backend)
- Run early-stage **architectural reviews** and verify that code follows **SOLID** principles and the team's coding standards
- Coordinate **API** interaction between frontend and backend, ensuring efficient two-way communication
- Leverage **WebGL**/**WebGPU** for high-performance **3D graphics**, writing custom shaders
- Designed and implemented an **Effect Compositor** system, integrating it into the **ECS** (**Entity-Component-System**). This significantly streamlined the creation of complex graphical effects and composite scenes.
- Implemented visual debugging tools that simplify the development process for junior engineers, accelerate feature delivery, and significantly reduce bugs.

Senior Frontend Engineer at Align Technology

July 2017 – January 2024

Developed a 3D editor for orthodontists (visualizing and editing jaw models). Contributed to UI, infrastructure, and WebSocket support.

- Built UI components using **React**, migrated project from ES6 to **TypeScript**
- Introduced **Redux** for state management, reducing technical debt
- Implemented runtime icon generation with **canvas 2D** for the 3D scene
- Created unit and integration tests (**Jest** + puppeteer)
- Managed backend communication via **WebSocket** (message queue, disconnection handling)
- Configured custom materials and shaders for **Three.js**, working with 3D math

Frontend Developer at Paragon Software

April 2015 – July 2017

Developed and maintained a single-page application (SPA) for the personal account system.

- Built the application with **AngularJS**, used responsive design techniques
- Exchanged data with the server via **RESTful API**
- Configured the build process using **Webpack**
- Developed a **Node.js** tool for localization (Excel → JSON)

Skills

JavaScript | TypeScript | C++ | ES6 | React | Redux | AngularJS | RxJS | WebGL | WebGPU | Three.js | Shaders | ECS | Node.js | RESTful API | WebSocket | Webpack | Jest | Puppeteer | Playwright | HTML | CSS/SASS | 3D Math

Education

Peoples' Friendship University of Russia (RUDN)

2006 – 2010

Bachelor's degree in General Physics

Peoples' Friendship University of Russia (RUDN)

2010 – 2012

Master's degree in General Physics

Languages

English — Fluent